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VEHICLE PRICE ANALYSIS & PREDICTIVE PRICING MODEL

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PROJECT SUMMARY

This project demonstrates how data can be used to support clear, reliable, and consistent pricing decisions by replacing subjective judgment with an objective, data-driven approach. The goal is not only to predict vehicle prices, but also to clearly explain why prices change and which factors truly influence value.

Most people already understand that larger vehicles tend to be more expensive and that newer vehicles typically command higher prices. What is often missing, however, is a precise understanding of how much these factors affect price. This project bridges that gap by developing a model that quantifies these relationships in measurable terms. Rather than relying on general statements such as “newer cars are worth more,” the model shows, for example, how much value is lost with each additional month of age, or how price changes as vehicle weight or engine power increases. By translating common assumptions into exact numbers, the analysis enables customers to move from broad expectations to specific, data-backed estimates that can be used confidently for pricing, comparison, and decision-making.

The analysis is based on vehicle sales data that includes final sale prices and key vehicle characteristics such as age, mileage, engine specifications, weight, and fuel type. Before building the model, the data was carefully reviewed and prepared to ensure accuracy and consistency. This preparation step is essential, as reliable data leads directly to more trustworthy results and stronger conclusions. A statistical pricing model was then developed to measure how each vehicle characteristic influences price while holding other factors constant. To ensure the model performs well in real-world situations, the data was divided into separate subsets for training, testing, and validation. This approach helps confirm that the model captures meaningful patterns rather than simply fitting historical data.

This project delivers both accurate predictions and clear, actionable insights. The same approach can be adapted to other pricing, forecasting, classification, or decision-support problems and tailored to different industries, datasets, and business objectives.

BUSINESS CONTEXT & PROBLEM DEFINITION

In many cases, pricing decisions are still based on manual estimates, personal experience, or general market intuition. While this approach may work at a high level, it often leads to inconsistent pricing, limited transparency, and difficulty explaining or justifying decisions to customers or other stakeholders. As pricing decisions scale across inventories, locations, or platforms, these limitations become increasingly significant.

This type of analysis is particularly useful for organizations and professionals who require consistent, transparent, and data-driven pricing or valuation decisions. Vehicle dealerships can use it to ensure fair and consistent pricing across their inventory, evaluate trade-ins objectively, and justify prices to customers using data rather than subjective judgment. Used-car marketplaces and online platforms can apply this approach to standardize pricing across large volumes of listings, reduce price dispersion, and build trust by offering clear and explainable price estimates. Financial institutions and lenders can rely on this analysis to assess collateral value, support loan and lease decisions, and improve risk evaluation by understanding how specific vehicle characteristics influence market value. Insurance companies can use the model to estimate vehicle values accurately for underwriting, claims assessment, and loss valuation, particularly in situations where consistency and transparency are critical. Fleet managers and rental companies can apply the same methodology to monitor depreciation, optimize asset replacement timing, and evaluate resale value using measurable and interpretable drivers of price.

The main question addressed by this project is: **How much does each vehicle characteristic, such as age, mileage, size, and engine specifications, affect price, and how can these effects be used to produce reliable, data-driven pricing decisions?**

DATA DESCRIPTION & PREPARATION

The analysis is based on a real-world dataset containing 1,000 vehicles from the same car brand, which allows for a consistent and controlled comparison across observations. Each record represents an individual vehicle and includes the final sale price along with key characteristics commonly used to describe and evaluate vehicles, such as age, mileage, horsepower, engine size, vehicle weight, number of doors, transmission type, and fuel type.

These variables were selected because they reflect the information typically available to dealerships, marketplaces, lenders, and insurers when evaluating a vehicle.

Before building the pricing model, the data underwent a careful preparation process to ensure accuracy and consistency. This included reviewing the dataset for missing or inconsistent values, checking that numerical variables fell within realistic ranges, and confirming that categorical information, such as fuel type and transmission, was recorded consistently.

Categorical variables were structured in a way that allows their individual impact on price to be measured objectively, while numerical variables were retained in their original units to preserve interpretability. This ensures that model outputs can be directly translated into practical insights, such as understanding how price changes with additional mileage or vehicle age.

By ensuring the data is clean, consistent, and well-structured, the analysis provides a solid foundation for building a pricing model.

Price	Age	Mileage	Fuel Type	Horsepower	Metallic	Automatic	CC	Doors	Weight
13,500	23	46,986	Diesel	90	1	0	2,000	3	1,165
13,750	23	72,937	Diesel	90	1	0	2,000	3	1,165
13,950	24	41,711	Diesel	90	1	0	2,000	3	1,165
14,950	26	48,000	Diesel	90	0	0	2,000	3	1,165
13,750	30	38,500	Diesel	90	0	0	2,000	3	1,170

Header of the dataset.

METHODOLOGY

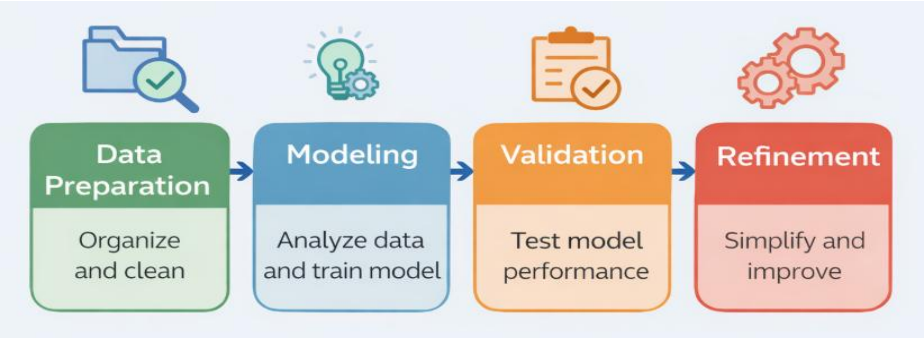
The objective of this analysis was to build a pricing model that is both accurate and easy to interpret, allowing pricing decisions to be supported by clear, measurable evidence. To achieve this, a structured and transparent modeling approach was followed.

First, vehicle prices were defined as the outcome of interest, while vehicle characteristics such as age, mileage, horsepower, engine size, weight, and fuel type were treated as explanatory factors. These variables were analyzed together to capture their combined effect on price, rather than evaluating each factor in isolation.

A regression-based modeling approach was used to quantify how each vehicle characteristic influences price while holding all other factors constant. This method was chosen because it provides clear, numerical estimates of impact that can be direct. Each model coefficient represents how much price is expected to change when a specific vehicle characteristic increases by one unit.

To ensure the model performs reliably in real-world scenarios, the dataset was divided into separate subsets for model development, testing, and validation. This step helps confirm that the model captures meaningful patterns rather than simply fitting historical data.

Model performance was evaluated using standard accuracy measures and by comparing predicted prices to actual market prices across all data subsets. This evaluation process helps identify potential overfitting and ensures that the model remains stable when applied to new, unseen vehicles.



Finally, the model was refined by removing variables that did not meaningfully improve predictive performance. This simplification step resulted in a more efficient and transparent model. The final model balances predictive strength with practical usability, making it well suited for real-world pricing and valuation applications.

ANALYSIS & MODEL RESULTS

To analyze vehicle prices and quantify the impact of individual vehicle characteristics, a multiple linear regression model was developed using the Python computer programming language. This modeling approach was selected because it provides both reliable predictions and clear interpretability, allowing pricing decisions to be supported by transparent, data-driven reasoning.

Definition: Multiple linear regression is a statistical method used to understand how one outcome depends on several factors at the same time. It describes the outcome as a combination of these factors, where each factor contributes its own effect. The method estimates how much the outcome changes when one factor changes, while all other factors are kept the same. This makes it possible to separate and measure the individual effect of each factor. Multiple linear regression is commonly used when the goal is to both predict values and understand how different variables influence the result.

The coefficients of the multiple linear regression model quantify how each vehicle characteristic affects price, assuming all other characteristics remain unchanged. This allows each factor's impact to be interpreted independently, providing a clear understanding of how different features contribute to vehicle value.

```
Model Coefficients:
  Age: -128.84
  Mileage: -0.02
  Horse Power: 51.55
  Metalic: 186.23
  Automatic: 588.75
  CC: -2.39
  Doors: -46.87
  Weight: 22.82
  Fuel Type_Diesel: 1644.75
  Fuel Type_Petrol: 1017.67

Intercept: -7906.96
```

The analysis shows that vehicle age is the strongest driver of price depreciation. For each additional month a vehicle is owned, the expected price decreases by approximately USD 128. This confirms that newer vehicles retain significantly higher value and that depreciation occurs steadily over time.

Mileage also has a negative effect on prices. Each additional mile drive reduces the estimated value by about USD 0.02, reflecting the impact of usage and wear on market value. Together, age and mileage account for a substantial portion of price variation and represent the core depreciation mechanisms.

Several characteristics contribute positively to vehicle prices. Horsepower increases in value, with higher-performance vehicles commanding higher prices in the market. Vehicle weight also has a positive effect, suggesting that larger or more robust vehicles tend to be priced higher, potentially due to perceived safety, build quality, or utility.

Engine size (CC) shows a small negative effect in this dataset, indicating that larger engines do not necessarily translate into higher prices once other factors are considered. This may reflect trade-offs related to fuel efficiency or operating costs within the same vehicle brand.

Fuel type plays a meaningful role in pricing. Compared to CNG vehicles, diesel vehicles are priced significantly higher, followed by petrol vehicles, even after accounting for age, mileage, and other characteristics. This indicates that fuel type adds independent value beyond basic vehicle specifications.

The model coefficients provide a transparent and interpretable breakdown of how vehicle characteristics combine to determine price. By translating each factor into a measurable monetary effect, the model enables clear, data-driven explanations of pricing decisions and supports consistent, objective vehicle valuation.

But how can this information be used to create an actual price prediction?

The regression coefficients on their own explain how each vehicle characteristic affects price, but their real value comes from combining them into a single pricing formula. The regression equation brings together all these individual effects and applies them at the same time to produce one estimated price.

$$\begin{aligned}\text{Price} = -7906.96 - 128.84 \times \text{Age} - 0.02 \times \text{Mileage} + 51.55 \times \text{Horsepower} + 186.23 \times \text{Metallic} \\ + 588.75 \times \text{Automatic} - 2.39 \times \text{Engine Size (CC)} - 46.87 \times \text{Doors} + 22.82 \times \text{Weight} \\ + 1644.75 \times \text{Fuel Type (Diesel)} + 1017.67 \times \text{Fuel Type (Petrol)}\end{aligned}$$

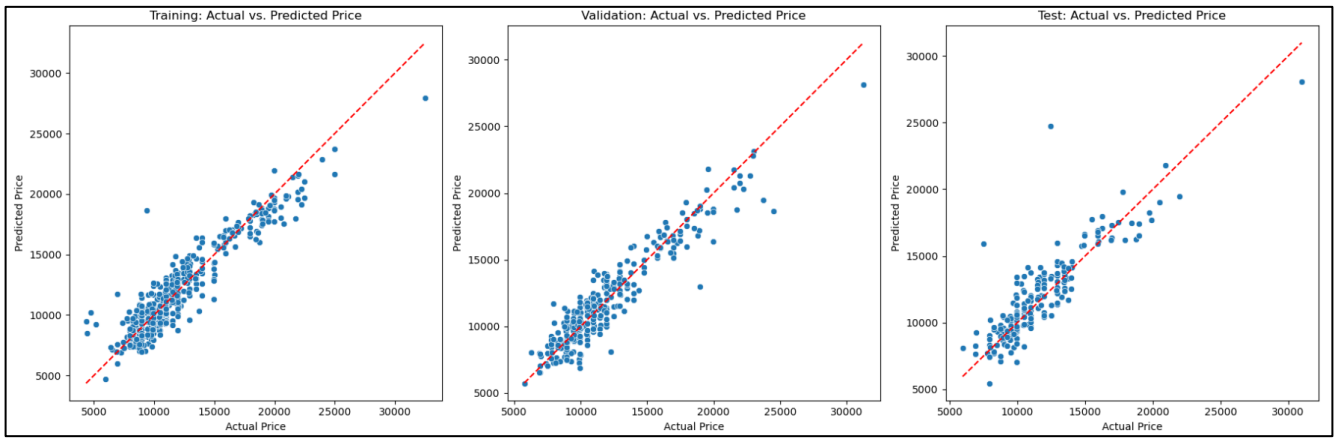
For example, considering a vehicle that is 24 months old with 40,000 miles of usage, 100 horsepower, an engine size of 1,600 cc, four doors, and a vehicle weighing 1,200 units. The vehicle does not have metallic paint and is not equipped with automatic transmission. It runs on diesel fuel, with CNG and petrol treated as alternative fuel categories.; the formula is:

Estimated Price = -7906.96 - 128.84*(24) - 0.02*(40000) + 51.55*(100) + 186.23*(0) + 588.75*(0) - 2.39*(1600) - 46.87*(4) + 22.82*(1200) + 1644.75*(1) + 1017.67*(0) = 18,373.15

Estimated Price = **USD 18,373.15**

MODEL PERFORMANCE

But how good is this predictive model? As described earlier, the dataset was split from the beginning into three separate parts: a training set, a validation set, and a test set. This approach ensures that model performance is evaluated not only on the data used to build the model, but also on new and previously unseen vehicles.



Model accuracy is first assessed visually using actual versus predicted price plots. In these charts, the horizontal axis shows the actual market price, while the vertical axis shows the price predicted by the model. Each point represents a vehicle, and the red dashed line indicates a perfect prediction. Points close to this line reflect accurate estimates. Across all three datasets, most points cluster near the reference line, indicating consistent predictive performance.

```
Model Performance:
  Training Set - RMSE: 1341.14, R-squared: 0.88
  Validation Set - RMSE: 1292.43, R-squared: 0.89
  Test Set - RMSE: 1647.93, R-squared: 0.75
```

Model performance is also quantified using R-squared and RMSE. The model explains approximately 88–89% of price variation in the training and validation sets, with average prediction errors of about USD 1,300 – 1650 USD. Both the plots and performance metrics confirm that the model provides reliable and consistent price estimates while generalizing well to unseen data.

MODEL SIMPLIFICATION

Can we simplify the model further? Are all variables truly necessary? While the initial regression model provides strong predictive performance, not every variable contributes equally to explaining vehicle price. In practice, including unnecessary variables can increase complexity. For this reason, the next step of the analysis focused on identifying whether the model could be simplified while maintaining predictive strength.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  Price   R-squared:                       0.880
Model:                            OLS   Adj. R-squared:                  0.878
Method:                 Least Squares   F-statistic:                     358.5
Date:                Thu, 18 Dec 2025   Prob (F-statistic):          7.31e-218
Time:                        15:17:05   Log-Likelihood:                -4310.1
No. Observations:                 500   AIC:                             8642.
Df Residuals:                     489   BIC:                             8689.
Df Model:                          10
Covariance Type:            nonrobust
====================================================================================
                       coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------------
const            -7906.9635   2205.764     -3.585      0.000   -1.22e+04   -3573.018
Age               -128.8440      5.287    -24.370      0.000    -139.232    -118.456
Mileage             -0.0184      0.003     -7.313      0.000      -0.023      -0.013
Horse Power         51.5470      8.356      6.169      0.000      35.130      67.964
Metalic            186.2274    131.019      1.421      0.156     -71.203     443.658
Automatic          588.7530    308.857      1.906      0.057     -18.098    1195.603
CC                  -2.3926      0.902     -2.652      0.008      -4.165      -0.620
Doors              -46.8703     69.485     -0.675      0.500    -183.396      89.655
Weight              22.8185      1.914     11.920      0.000      19.057      26.580
Fuel Type_Diesel  1644.7532    865.880      1.900      0.058     -56.552    3346.058
Fuel Type_Petrol  1017.6698    643.386      1.582      0.114    -246.473    2281.813
==============================================================================
Omnibus:                      108.919   Durbin-Watson:                   2.099
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              624.440
Skew:                          -0.807   Prob(JB):                    2.54e-136
Kurtosis:                       8.231   Cond. No.                     2.69e+06
==============================================================================
```

To evaluate this, a statistical version of the regression model was estimated using Ordinary Least Squares (OLS). This approach provides additional diagnostic information, including statistical significance measures that help assess whether each variable meaningfully contributes to price prediction when all other variables are considered. Variables with weak or inconsistent contributions were flagged for potential removal (where p-value > 0.05).

Based on this analysis, several variables, such as metallic paint, transmission type, number of doors, and fuel type, were found to add limited explanatory value once the main drivers of price were already included. These variables were therefore removed using a elimination procedure, which iteratively retains only predictors that meet a clear statistical significance threshold.

The resulting simplified model retains only the five most important drivers of vehicle price: Age, Mileage, Horsepower, Engine size (CC), and Vehicle weight.

After re-estimating the model using only these variables, performance remained nearly unchanged. The simplified model explains approximately 88–89% of price variation on the training and validation datasets and maintains strong performance on the test dataset, with only a minor reduction in accuracy. This confirms that the removed variables did not materially improve predictive power.

FINAL PRICING MODEL

After simplifying the original regression model, the final pricing model focuses only on the five most important drivers of vehicle price. These variables consistently explain most of the price variation while keeping the model easy to understand, maintain, and apply in practice.

```
Model Coefficients with Selected Features:
  Age: -128.14
  Mileage: -0.02
  Horse Power: 44.60
  CC: -1.62
  Weight: 23.51
```

Vehicle age has the strongest negative effect, with each additional month reducing the estimated price by approximately USD 128, reflecting steady depreciation over time. Mileage also decreases value, with each additional mile lowering the estimated price by about USD 0.02, which accumulates into a meaningful impact as usage increases. Horsepower contributes positively to price, increasing the estimated value by roughly USD 44.60 per unit, indicating that higher-performance vehicles are generally valued more highly. Engine size (CC) has a small negative effect, suggesting that larger engines do not necessarily add value once performance and other factors are considered, possibly reflecting efficiency preferences within the same brand. Vehicle weight has a positive effect on price, with heavier vehicles commanding higher estimated values, likely due to perceived build quality or utility. The intercept represents the baseline level of the model before vehicle characteristics are applied and ensures that the pricing equation produces accurate estimates.

Final Pricing Formula:

Price = -8079.33 - 128.14*Age - 0.02*Mileage + 44.60*Horsepower - 1.62*CC + 23.51*Weight

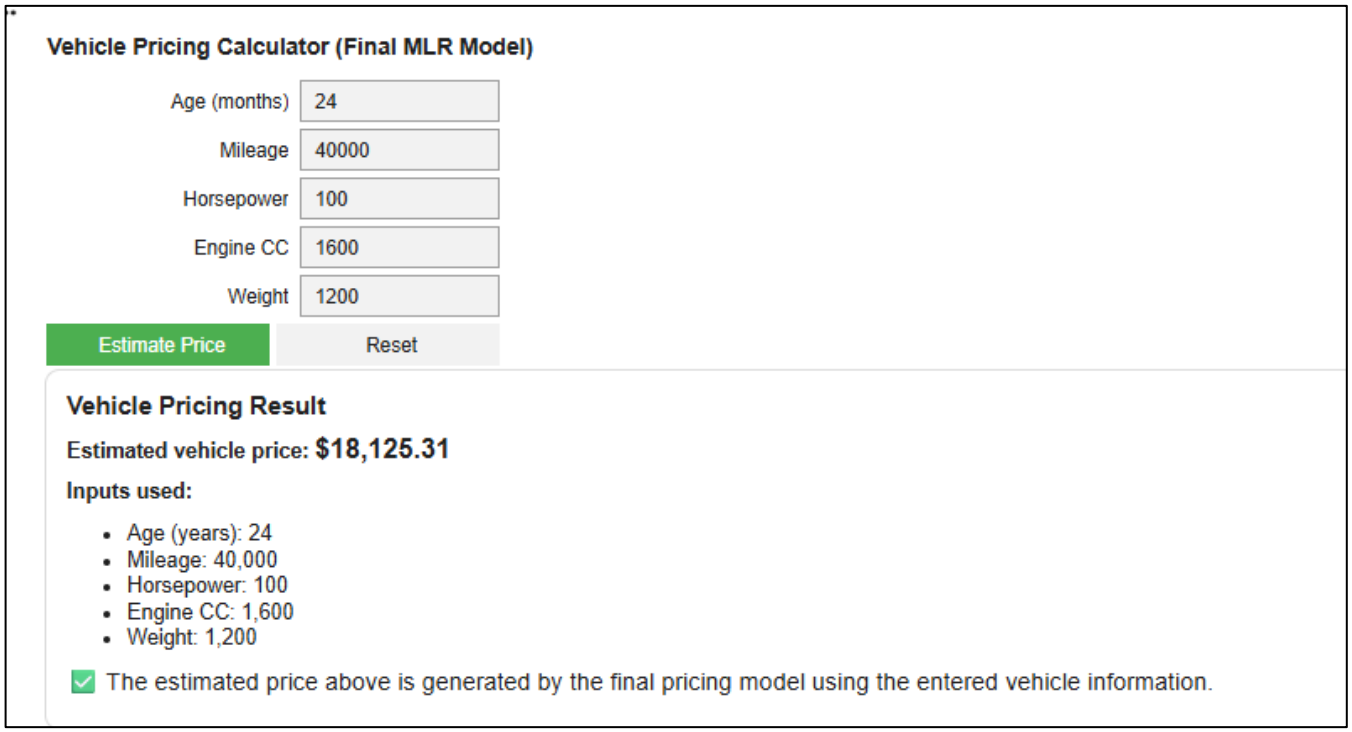
Example:

For a car with an age of 24 months, mileage of 40,000 miles, horsepower of 100, engine size of 1,600 cc, and weight of 1,200, the estimated price is calculated as: $-8{,}079.33 - 128.14 \times 24 - 0.02 \times 40{,}000 + 44.60 \times 100 - 1.62 \times 1{,}600 + 23.51 \times 1{,}200 =$ USD 18,125

FROM MODEL TO PRACTICAL PRICING TOOL

While the final pricing model provides a clear mathematical representation of how vehicle characteristics influence price, its real value lies in how it can be applied in practice. To translate the model into a usable decision-support tool, the pricing equation was implemented in an interactive vehicle pricing calculator.

This interface allows users to enter basic vehicle information and instantly receive an estimated market price generated by the final pricing model. By embedding the model logic behind a simple, user-friendly form, the tool enables customers and stakeholders to benefit from the analysis without needing to understand the underlying statistical methodology. The calculator demonstrates how a data-driven pricing model can be operationalized into a practical application that supports transparent, consistent, and objective pricing decisions in real-world settings.



BUSINESS IMPLICATIONS & RECOMMENDATIONS

This analysis shows that vehicle pricing can be made consistent, explainable, and scalable by relying on a few measurable factors rather than subjective judgment. Age and mileage drive depreciation in a predictable way, while performance and vehicle size contribute positively to value. By quantifying these effects, pricing decisions become easier to justify, communicate, and replicate across vehicles and locations.

Recommendation 1: Use the model as a pricing benchmark.
The final pricing model can be used as a baseline estimate for vehicle value. This allows businesses to quickly identify vehicles that may be underpriced or overpriced relative to their characteristics and market behavior. Prices can then be adjusted strategically rather than based on intuition alone.

Recommendation 2: Standardize pricing and valuation processes.
Because the model relies on a limited set of core inputs, it can be applied consistently across inventory, locations, or platforms. This reduces pricing variability between decision-makers and increases transparency for customers, partners, and internal teams.

Recommendation 3: Improve negotiations and customer communication.
The model provides a clear, data-backed explanation for price differences. This makes it easier to justify pricing during negotiations, trade-in evaluations, or financing discussions by pointing to measurable factors such as age, mileage, and performance.

Recommendation 4: Operationalize the model through an interactive pricing interface.
By embedding the final pricing model into a user-friendly calculator, the analysis can be directly applied in day-to-day decision-making. The interface allows users to enter vehicle characteristics and instantly receive a price estimate, making the model accessible to non-technical stakeholders. This reduces reliance on manual calculations, increases consistency, and enables faster, more transparent pricing decisions in customer-facing or operational settings.

Recommendation 5: Monitor depreciation and asset value over time.
The quantified depreciation effects can be used to track how vehicle value changes over time and usage. This supports better decisions around inventory turnover, resale timing, and asset management.

Recommendation 6: Extend the approach to other use cases.
The same framework can be adapted to other brands, vehicle categories, or valuation problems. With additional data, the model can be customized to support forecasting, risk assessment, or automated pricing tools.

This project demonstrates how a simple pricing model can support better business decisions, reduce uncertainty, and increase confidence in vehicle pricing and valuation, without requiring complex systems.

Appendix (Technical & Codes used for this analysis)

```python
from sklearn.model_selection import train_test_split

# Defining the target variable and predictor features
target = 'Price'
features = df.drop(target, axis=1)
target_variable = df[target]

# Splitting the data to reserve 20% for final model testing
X_train, X_test, y_train, y_test =  train_test_split(features, target_variable, test_size=0.2, random_state=42)

# Further splitting the remaining data into training (50%) and validation (30%)
X_train, X_val, y_train, y_val = train_test_split(X_train, y_train, test_size=0.375, random_state=42) # 0.375 * 0.8 = 0.3

# Verifying the dimensions of each dataset split
print("Training set shape:", X_train.shape, y_train.shape)
print("Validation set shape:", X_val.shape, y_val.shape)
print("Test set shape:", X_test.shape, y_test.shape)
```

```
Training set shape: (500, 10) (500,)
Validation set shape: (300, 10) (300,)
Test set shape: (200, 10) (200,)
```

```python
# Displaying model coefficients to interpret the direction and strength of each predictor
print("\nModel Coefficients:")
for feature, coef in zip(X_train.columns, model.coef_):
    print(f"  {feature}: {coef:.2f}")

# Displaying the intercept term (baseline prediction when all predictors are zero)
print(f"\nIntercept: {model.intercept_:.2f}")
```

```
Model Performance:
  Training Set - RMSE: 1341.14, R-squared: 0.88
  Validation Set - RMSE: 1292.43, R-squared: 0.89
  Test Set - RMSE: 1647.93, R-squared: 0.75

Model Coefficients:
  Age: -128.84
  Mileage: -0.02
  Horse Power: 51.55
  Metalic: 186.23
  Automatic: 588.75
  CC: -2.39
  Doors: -46.87
  Weight: 22.82
  Fuel Type_Diesel: 1644.75
  Fuel Type_Petrol: 1017.67

Intercept: -7906.96
```

```python
# Constructing and displaying the final regression equation
# Extracting the intercept and estimated coefficients from the trained model
intercept = model.intercept_
coefficients = model.coef_
feature_names = X_train.columns

# Initializing the regression formula with the intercept term
formula = f"Price = {intercept:.2f}"

# Appending each predictor and its coefficient to form the full equation
for feature, coef in zip(feature_names, coefficients):
    if coef >= 0:
        formula += f" + {coef:.2f} * {feature}"
    else:
        formula += f" - {-coef:.2f} * {feature}"

# Displaying the complete regression formula
print("Regression Formula:")
print(formula)
```

```
Regression Formula:
Price = -7906.96 - 128.84 * Age - 0.02 * Mileage + 51.55 * Horse Power + 186.23 * Metalic + 588.75 * Automatic - 2.39 * CC - 46.87 * Doors + 22.82 * Weig
ht + 1644.75 * Fuel Type_Diesel + 1017.67 * Fuel Type_Petrol
```

```python
# Fitting an OLS regression model with statsmodels to obtain detailed statistical output
# This includes p-values, confidence intervals, and overall significance measures

X_train_sm = X_train.copy()
for col in ['Fuel Type_Diesel', 'Fuel Type_Petrol']:
    if col in X_train_sm.columns and X_train_sm[col].dtype == bool:
        X_train_sm[col] = X_train_sm[col].astype(int)

# Adding an intercept term (constant) so the model estimates a baseline price level
X_train_sm = sm.add_constant(X_train_sm)

# Training the OLS regression model on the training dataset
model_sm = sm.OLS(y_train, X_train_sm).fit()

# Displaying a full statistical summary (coefficients, p-values, and fit statistics)
print(model_sm.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  Price   R-squared:                       0.880
Model:                            OLS   Adj. R-squared:                  0.878
Method:                 Least Squares   F-statistic:                     358.5
Date:                Fri, 19 Dec 2025   Prob (F-statistic):          7.31e-218
Time:                        12:48:53   Log-Likelihood:                -4310.1
No. Observations:                 500   AIC:                             8642.
Df Residuals:                     489   BIC:                             8689.
Df Model:                          10
Covariance Type:            nonrobust
=====================================================================================
                        coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------------
const             -7906.9635   2205.764     -3.585      0.000   -1.22e+04   -3573.018
Age                -128.8440      5.287    -24.370      0.000    -139.232    -118.456
Mileage              -0.0184      0.003     -7.313      0.000      -0.023      -0.013
Horse Power          51.5470      8.356      6.169      0.000      35.130      67.964
Metalic             186.2274    131.019      1.421      0.156     -71.203     443.658
Automatic           588.7530    308.857      1.906      0.057     -18.098    1195.603
CC                   -2.3926      0.902     -2.652      0.008      -4.165      -0.620
Doors               -46.8703     69.485     -0.675      0.500    -183.396      89.655
Weight               22.8185      1.914     11.920      0.000      19.057      26.580
Fuel Type_Diesel   1644.7532    865.880      1.900      0.058     -56.552    3346.058
Fuel Type_Petrol   1017.6698    643.386      1.582      0.114    -246.473    2281.813
==============================================================================
Omnibus:                      108.919   Durbin-Watson:                   2.099
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              624.440
Skew:                          -0.807   Prob(JB):                    2.54e-136
Kurtosis:                       8.231   Cond. No.                     2.69e+06
==============================================================================
```

```python
# Automating variable selection using backward elimination (p-value based)

def backward_elimination(X, y, significance_level=0.05):
    """
    Performs backward elimination to select features based on p-values.

    Args:
        X (pd.DataFrame): DataFrame of predictor variables.
        y (pd.Series): Series of the target variable.
        significance_level (float): The alpha level for statistical significance.

    Returns:
        list: A list of the selected feature names.
    """
    features = X.columns.tolist()
    while (len(features) > 0):
        # Create a copy and convert boolean columns to integers for statsmodels compatibility
        X_be = X[features].copy()
        for col in X_be.columns:
            if X_be[col].dtype == bool:
                X_be[col] = X_be[col].astype(int)

        X_train_be = sm.add_constant(X_be)
        model_be = sm.OLS(y, X_train_be).fit()
        p_values = model_be.pvalues[1:] # Exclude constant's p-value
        max_p_value = max(p_values)
        if max_p_value > significance_level:
            redundant_feature = p_values.idxmax()
            features.remove(redundant_feature)
        else:
            break
    return features
```

```python
# Running backward elimination on the training dataset
selected_features = backward_elimination(X_train, y_train)

# Reporting the final set of retained predictors
print("Selected features after backward elimination:", selected_features)

# Updating training, validation, and test sets to include only the selected predictors
X_train_selected = X_train[selected_features]
X_val_selected = X_val[selected_features]
X_test_selected = X_test[selected_features]

# Verifying the updated dataset dimensions after feature selection
print("\nShape of training data with selected features:", X_train_selected.shape)
print("Shape of validation data with selected features:", X_val_selected.shape)
print("Shape of test data with selected features:", X_test_selected.shape)
```

```
Selected features after backward elimination: ['Age', 'Mileage', 'Horse Power', 'CC', 'Weight']

Shape of training data with selected features: (500, 5)
Shape of validation data with selected features: (300, 5)
Shape of test data with selected features: (200, 5)
```

So let's run the model again with fewer predictors and see how it performs.

```python
# Re-training the linear regression model using the selected subset of predictors
# Initializing the regression model for the reduced feature set
model_selected = LinearRegression()

# Fitting the model using training data with selected predictors only
model_selected.fit(X_train_selected, y_train)

# Generating predictions across training, validation, and test splits
y_train_pred_selected = model_selected.predict(X_train_selected)
y_val_pred_selected = model_selected.predict(X_val_selected)
y_test_pred_selected = model_selected.predict(X_test_selected)

# Evaluating model accuracy using MSE and RMSE (RMSE is in the original price units)
train_mse_selected = mean_squared_error(y_train, y_train_pred_selected)
val_mse_selected = mean_squared_error(y_val, y_val_pred_selected)
test_mse_selected = mean_squared_error(y_test, y_test_pred_selected)

# Calculate RMSE for the new model
train_rmse_selected = np.sqrt(train_mse_selected)
val_rmse_selected = np.sqrt(val_mse_selected)
test_rmse_selected = np.sqrt(test_mse_selected)

# Calculating R-squared to measure explanatory power on each split
train_r2_selected = r2_score(y_train, y_train_pred_selected)
val_r2_selected = r2_score(y_val, y_val_pred_selected)
test_r2_selected = r2_score(y_test, y_test_pred_selected)

# Reporting performance to compare the reduced model against the full model
print("Model Performance with Selected Features:")
print(f"  Training Set - RMSE: {train_rmse_selected:.2f}, R-squared: {train_r2_selected:.2f}")
print(f"  Validation Set - RMSE: {val_rmse_selected:.2f}, R-squared: {val_r2_selected:.2f}")
print(f"  Test Set - RMSE: {test_rmse_selected:.2f}, R-squared: {test_r2_selected:.2f}")
```

```python
# Displaying coefficients to interpret how each retained predictor impacts the price estimate
print("\nModel Coefficients with Selected Features:")
for feature, coef in zip(X_train_selected.columns, model_selected.coef_):
    print(f"  {feature}: {coef:.2f}")

# Displaying the intercept term for the reduced model
print(f"\nIntercept: {model_selected.intercept_:.2f}")
```

```
Model Performance with Selected Features:
  Training Set - RMSE: 1354.25, R-squared: 0.88
  Validation Set - RMSE: 1282.72, R-squared: 0.89
  Test Set - RMSE: 1673.41, R-squared: 0.74

Model Coefficients with Selected Features:
  Age: -128.14
  Mileage: -0.02
  Horse Power: 44.60
  CC: -1.62
  Weight: 23.51

Intercept: -8079.33
```